# Developing Robust ASR for Autism-Focused Reading Comprehension Therapy

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**ABSTRACT**

Automatic Speech Recognition (ASR) systems, which use machine learning to transcribe speech, have transformed how humans interact with technology. However, despite recent advances, they remain unreliable for individuals with disordered or atypical speech. This limits the feasibility of digital speech therapies that rely on verbal communication. Many neurodiverse learners depend on verbal interventions such as Lindamood-Bell's Visualization & Verbalization, which develop comprehension through guided verbalization with visual prompts. However, ASR inaccuracies make such therapies impractical to scale digitally. This project introduces a novel ASR framework that combines domain-specific fine-tuning with multimodal adaptation to model speech patterns that existing systems overlook. I propose fine-tuning a pre-trained ASR model using low-rank adaptation (LoRA) on a dataset of disordered speech and integrating multimodal decoder biasing to align recognition with concurrent visual cues—an innovative approach in therapeutic speech modeling. The model will be evaluated using word error rate. By achieving accurate and context-aware recognition of atypical speech, this work can unlock scalable, low-cost digital therapies for individuals who currently lack access to in-person treatment—bridging the gap between cutting-edge AI and equitable speech therapy delivery worldwide.

**INTRODUCTION**

Up to 73% of the autistic community excel in decoding words but struggle with reading comprehension (Kritsotakis et al., 2025). Dual-coding theory posits that cognition requires an

intertwining of two pathways—a verbal pathway for reading and a non-verbal pathway for non-verbal information in the form of mental imagery (Paivio, 2014). Reading comprehension interventions based on this theory function by pairing visual information with verbal descriptions: children are shown curated visual prompts and guided towards describing them in detail. Their responses are then scaffolded through structured prompting, eventually developing into complete and coherent answers. One well-known intervention for such challenges is Visualization and Verbalization (V/V), a program based on the dual-coding theory of cognition. The efficacy of V/V for Autism Spectrum Disorder (ASD) was investigated by Murdaugh et al., (2015) and Beckerson et al., (2024). Their studies have shown that the therapy significantly improved comprehension and increased activations in areas of the brain associated with language and visual processing. However, few therapists are trained to deliver it, which makes the program difficult to access, drives up cost, and delays care.

To improve accessibility, I propose ADI/O, an AI-assisted therapeutic tool that operationalizes dual-coding based therapy. ADI/O is designed to strengthen the connection between visual and verbal understanding through interactive questioning and natural speech. A core component of this therapy is verbalization, which enables children to express themselves freely and supports the development of cognitive skills (Schunk, 1986). To replicate the authentic interactive experience central to dual-coding based therapy, the system relies on ASR. The challenge, however, is that the population of children likely to benefit from ADI/O is also likely to have speech or motor disorders such as slurring, atypical articulation, and inconsistent rhythm. Under these non-optimal conditions, ASR performance is significantly degraded (Tobin et al., 2024). Therefore, to deliver effective therapy, an accurate and reliable ASR is critical.

To improve ASR performance on disordered speech, several strategies such as personalization, data supplementation, and multimodal inputs have been explored. Each method has shown promise independently, but has not achieved the near-perfect performance required for therapeutic applications. Personalization, for example, adapts models to an individual speaker's characteristics. Studies such as Tobin et al. (2021) have demonstrated that just 3–4 minutes of additional speech could reduce word error rate (WER) from 31% to 15% for some users.

Data supplementation techniques provide additional context-specific data to improve performance, with Project Euphonia from Google serving as a widely used dataset for disordered speech. Fine-tuning ASR models on such datasets, as explored in Mutjaba et al. (2025), has also yielded improvements, reducing WER from 33% to 18%. Another promising approach is multimodal-decoder biasing, which incorporates visual context into ASR systems. Visual cues—such as images, scenes, and videos—can provide additional semantic context for phonetically ambiguous speech. This is achieved by first converting both the ASR model's hypotheses and visual input into embeddings within a shared semantic space. The cosine similarity between these embeddings indicates how well the transcription aligns with the visual scene. When the similarity score is combined with the ASR model's scores, the decoder biases towards hypotheses that are both phonetically and visually consistent. Prior work, including Hu et al. (2024), demonstrates that leveraging visual embeddings can significantly improve transcription in context-rich scenarios.

Existing approaches improve ASR accuracy broadly, but are not optimized for tasks that involve describing visual scenes, where language is tightly linked to imagery. My approach

explicitly incorporates image embeddings during model inference, allowing the model to use visual context to transcribe scene-based speech in dual-coding based therapy more accurately.

**SOLUTION**

The platform combines five key components: an image bank, a gold-standard description generator, a multimodal ASR model, an intelligent therapy scaffolding module, and a user evaluation module.

**1. Image Bank:** A curated collection of 100 high-quality, visually clear images rich in descriptive features yet simple enough to avoid overwhelming users. Images, such as the boy holding an umbrella (Fig. 1), will be used.

**2. Gold-Standard Description Generation Module:** This module uses an LLM to create detailed, developmentally appropriate image descriptions. These are organized by key structure words (e.g., what, where, who, color, size, mood) and serve as reference standards for evaluating user responses and guiding therapy.

**3. ASR Transcription Module**: An ASR model with near-perfect accuracy, as measured in Word Error Rate (WER). To achieve this, I will fine-tune Whisper on the Project Euphonia dataset and integrate a visual context pipeline using a joint image-text embedder, Contrastive Language-Image Pre-training (CLIP).

Whisper is OpenAI's advanced ASR system for accurate speech-to-text transcription. As it is trained on a large, multilingual dataset, it performs robustly across different accents and dialects. This project uses Whisper Small for its balance of speed and accuracy.

To fine-tune Whisper for disordered speech, I will use Google's Project Euphonia dataset, which includes over one million utterances and 1,300 hours of speech from speakers with diverse speech disorders. Its scale, quality, and consistency make it ideal for training ASR systems that generalize across disorder types, accents, and speaking conditions (Tobin et al., 2024).

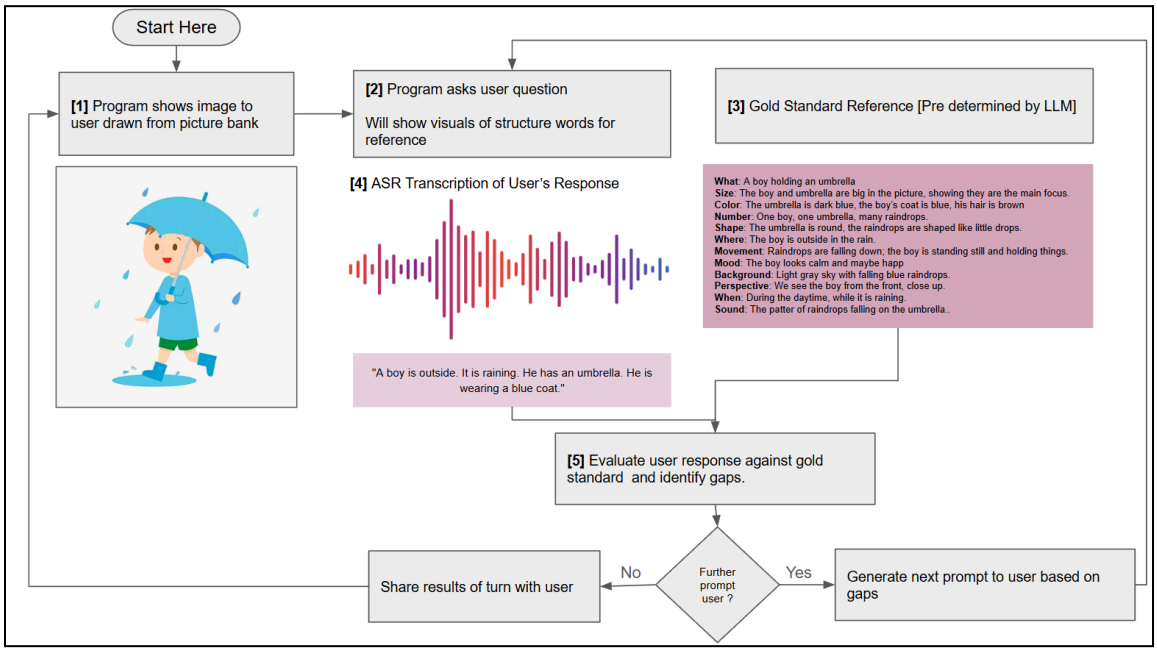


*Figure 1: User flow of ADI/O*

I will use CLIP to incorporate visual input into ASR. CLIP is a multimodal model that learns to map images and text in a shared embedding space by learning their semantic relationships. This can then be used to determine which text best describes a given image.

**3.1 Preprocessing:** During preprocessing, I will clean audio files to remove background noise and microphone artifacts as they disproportionately affect atypical speech. By normalizing audio formats (sampling rate, bit depth), I will ensure consistency across speakers. I will then apply data augmentation techniques such as controlled noise

injection, pitch shifting, and time-stretching to expand the dataset and improve overall model performance.

**3.2 Modeling:** I will apply Low Rank Adaptation (LoRA), as explored by Yu et al. (2023), to specialize existing speech recognition models. LoRA enables parameter-efficient fine-tuning by freezing the original model weights and learning only a small set of additional parameters that capture domain-specific adaptations. In their study, Yu et al. introduced LoRA as an efficient method for adapting large pre-trained language models for re-scoring in speech recognition. The technique involves inserting small, trainable, low-rank matrices into the model, while keeping most parameters frozen. This significantly reduces memory and computational requirements yet maintains high performance, making LoRA practical for deployment in resource-constrained settings. Training will use an 80–20 split for the training and test sets, with a separate validation set used during training.

Simultaneously, the ASR model generates multiple candidate transcriptions of the audio input. These candidate outputs are also converted into text embeddings using the text encoder of CLIP. Next, I will compare each text embedding with image embedding using cosine similarity, which indicates how closely the semantic meaning of the ASR candidate matches the visual context. I will rescore each ASR candidate based on the original score provided by the speech model, and final outputs are determined by spoken context and visual cues. This improves the overall performance in noisy and ambiguous conditions. The approach is visually represented in Fig. 2.

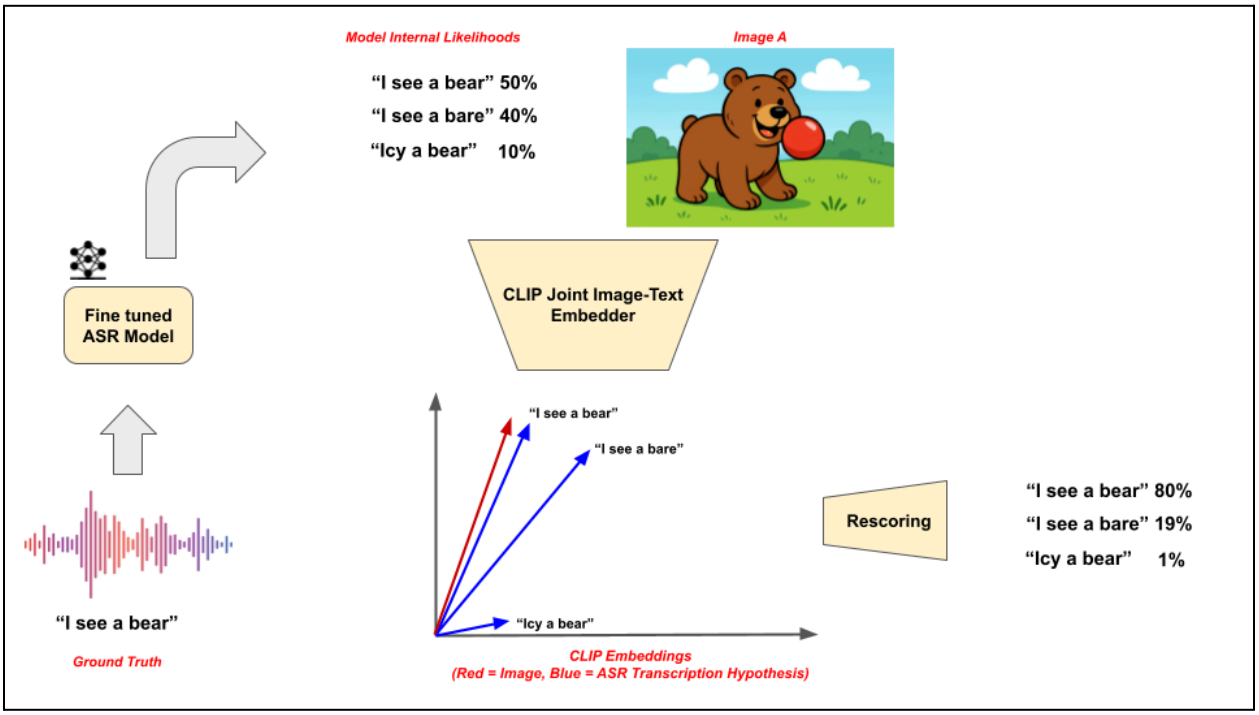


*Figure 2: Multimodal decoder-biasing pipeline*

**3.3 Evaluation** First, the curated image set will be provided to an LLM, which will generate a set of structured comprehension questions and gold-standard answers for each image. These questions will be grounded in the "structure word" framework used in V/V. For instance, if an image depicts "a child carrying an umbrella in the rain," the LLM might produce questions such as "Who is in the image?", "What is the person doing?", and "What color is the umbrella?", followed by ideal descriptive answers like "A child is carrying a blue umbrella." A target speaker with disordered speech will then record these answers, producing an aligned validation dataset of image and audio pairs. This dataset will represent the type of speech encountered in therapeutic use and will serve as the benchmark for evaluating model performance.

The validation process will proceed in three stages. First, I will establish a baseline by testing the validation dataset against the unmodified Whisper model and calculating its word error rate (WER). Next, I will fine-tune Whisper on the Project Euphonia dataset and re-evaluate performance using the same validation set to measure the improvement attributable to fine-tuning. Finally, I will integrate CLIP-based image embeddings at inference time so that the ASR output can be evaluated jointly with the visual context. The expectation is that incorporating image grounding will reduce ambiguities in descriptive speech—for example, distinguishing between "bear" and "bare" in disordered articulation—and further lower WER. This progressive validation approach ensures that each enhancement (fine-tuning and multimodal integration) is evaluated quantitatively.

**4. Therapeutic Scaffolding Module**: An LLM will generate scaffolded questions using key structure words to guide users toward detailed, comprehensive descriptions. These questions and a model example response help break down complex scenes, enabling children to understand them better and articulate visual content more effectively.

**5. User Performance Evaluation Module:** User responses will be evaluated against the gold standard using four main criteria: Omission, Extra Details, Detail and Clarity, and Linguistic Accuracy. A refinement of this approach will be the focus of future work.

## RISKS AND MITIGATION

**Overfitting & Modality Collapse:** I'll address overfitting by experimenting with multiple data splits (e.g., 80/20, 60/40) and tuning hyperparameters. Since the model may favor one modality

(like visuals) over another, I'll balance both speech and visual cues by regulating contextual influence.

**Costly Finetuning**: Training large ASR models on Project Euphonia can be expensive, so I'll use parameter-efficient approaches like LoRA and utilize free or cloud computing resources to optimize costs and performance.

**CLIP Not Suited for Conversation**: CLIP's text encoder, built for written captions, may not fit conversational transcriptions. To improve this, I'll preprocess transcription by cleaning up syntax and normalizing grammar. If this does not work, I could fine-tune the encoder on speech-based or conversational datasets.

## PROJECT LOGISTICS

The project plan consists of five phases. I am currently working on Phase 1 and plan to complete it by January 2026. Phases 2 through 4 will take 24 weeks (6 months). Phase 5 planned as future work.

**Phase 1 - Imagery Dataset and Visual Framework:** I will assemble an image dataset with metadata such as scene type and complexity. I will finalize a structure-word mapping that links each image to different question types. I will validate the dataset for coverage, clarity, and alignment with the verbal prompts. By the end, I will have a well-curated visual dataset.

**Phase 2 - Audio Dataset Preparation and ASR Modeling:** In this phase, I will clean, normalize and label all audio files with metadata such as speaker type and recording conditions. Next, I will collect some human samples of atypical speech and run it against the baseline Whisper model and measure WER. I will then fine tune the model using LoRA using my

processed audio dataset and measure the improvement in accuracy. I will conclude this phase when I have documented measured gains in transcription quality.

**Multimodality Development:** My goal for this phase is to create a system that integrates visual cues with transcribed speech. First, I will generate CLIP embeddings for each image and combine them with their corresponding ASR transcript candidates. I will test different fusion coefficients to see how visual context changes transcript ranking and whether it improves accuracy on speech. The phase will end when I choose the coefficient that produces the most consistent gains.

**Phase 3 - Therapy Application Prototype:** I will combine all the elements into a working prototype. I will define how the system presents images, asks questions, and delivers automated feedback. I will test the prototype for stability and basic interaction flow. This phase will end once I have a functional version ready for full evaluation.

**Phase 4 - Final Testing1 and Reporting:** I will run end to end tests to measure final transcription accuracy, multimodal gains, and overall interaction quality. I will gather example transcripts, scoring comparisons, and key error cases to support my results. I will present these findings with clear tables and visualizations. This phase will end with a complete written report.

**Phase 5 (Future Work): Extended Development and Research** - I will address the remaining foundational challenges, including enabling AI to reason about language with therapist-like precision and defining measurable comprehension indicators. Additional features such as gamification, personalization, and detailed progress reporting will be developed.

## RESOURCES NEEDED

**Mentorship:** My project lies at the intersection of speech recognition technology, cognitive neuroscience, and the use of AI in therapeutic settings. I believe the mentorship I require for this cross-functional project lies at MIT. Specifically, I see a strong overlap between my project objectives and the Spoken Language Systems Group at MIT CSAIL, which specializes in advancing speech recognition and understanding. MIT Media Lab's "Life with AI" initiative focuses on designing human-centered AI systems that enhance trust, equity, and meaningful learning outcomes. Finally, the McGovern Institute for Brain Research, focuses on cognitive neuroscience.

**Funding:** I am seeking funding of $1000, which will be allocated as follows:

| **Providers** | **Description** | **Cost** |
|---|---|---|
| Finetuning costs | Compute time and storage for fine-tuning the Whisper model. An estimated 25–30 GPU hours on rented cloud instances. | $200 |
| Domain Costs | Subscription costs for the website domain | $30 |
| Hosting Costs | Subscription costs for hosting a website | $70 |
| AWS Bedrock | Subscription costs for developing and deploying an AI application and accessing ChatGPT and Whisper APIs | $600 |
| Computation Power | Approx. 20 hours of GPU/CPU time for running model inference, evaluation on test datasets, and multimodal rescoring experiments. | $100 |
| **Total** | | **$1000** |

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